

The exact one-dimensional Taylor tree for a monomial perturbation

Claude, with Daniel Murfet

2026-09-06

Abstract

The simplest nontrivial instance of `thm:TaylorTree`, made exact. With $d = 1$, $\eta \equiv 1$ and $\xi(u) = \xi_0 + cu^m$, the chart standard integral over $(0, b]$ equals the convergent series $\sum_p \frac{(\beta c)^p}{p!} \int_0^b u^{h+mp} (\sqrt{n} u^k)^p e^{-\beta n u^{2k} + \beta \sqrt{n} u^k \xi_0} du$, and each term is $(2k)^{-1} n^{-(h+mp+1)/(2k)} S_{(h+mp+1)/(2k)+p/2}(\xi_0)$ minus an exponentially small tail. The exponents $(h+1)/(2k) + mp/(2k)$ run along the arithmetic progression of the candidate exponent set $\Lambda(h, k)$ — the tree's $n^{-\mu}$ structure, with fluctuation-function coefficients, in one dimension. Zero sorry/axiom.

1 Setting

Units 22–25 supplied the analytic steps of the one-dimensional Taylor-tree pipeline: the kernel identity, insertions as a -derivatives, Gaussian domination, and the boundary tail. What remained was the series interchange for $e^{\beta \sqrt{n} u^k (\xi(u) - \xi(0))}$. For a single monomial $\xi(u) - \xi(0) = cu^m$ this is one exponential series, and on the bounded chart the interchange is dominated convergence against a constant multiple of $u^h e^{-\beta n u^{2k} + \beta \sqrt{n} u^k \xi_0}$ — exactly the route the paper takes before pushing each term to the half-line.

2 The things formalised

```
theorem standardIntegral1D_monomial_hasSum ( n c b : ) (h k m : )
  (h : 0 < ) (hb : 0 < b) :
  HasSum (fun p : => (*c)^p / p.factorial
    * u in Ioc 0 b, u^(h + m*p) * (Real.sqrt n * u^k)^p * exp(-
      n u^{2k} + sqrt n u^k ))
    ( u in Ioc 0 b, u^h * exp(- n u^{2k} + sqrt n u^k ( + c u^m)))

theorem standardIntegral1D_monomial_expansion ... (hn : 0 < n) (hk : 0 < k) :
  HasSum (fun p : => (*c)^p / p.factorial
    * ((1/(2k)) * n^(-(h+mp+1)/(2k))) * fluctuation ((h+mp+1)/(2k) + p/2)
      - u in Ioi b, u^(h + m*p) * (sqrt n u^k)^p * exp(...))
    ( u in Ioc 0 b, u^h * exp(- n u^{2k} + sqrt n u^k ( + c u^m)))
```

3 Proof strategy

Following the `fluctuation_log_generating` template, apply `hasSum_integral_of_dominated_convergence` on $\mu = \lambda|_{(0,b]}$ with $F_p(u) = \frac{(\beta c)^p}{p!} u^{h+mp} (\sqrt{n} u^k)^p e(u)$. The key rewriting is $F_p(u) = u^h e(u) \cdot x^p / p!$ with $x = \beta \sqrt{n} u^k c u^m$

(mul_pow, pow_mul, pow_add), so that (i) the pointwise limit is $u^h e(u) e^x = f(u)$ by `NormedSpace.expSeries_div_hasSum_exp` and `HasSum.mul_left`, and (ii) the bound is $B^p/p! \cdot u^h e(u)$ with the constant $B = \beta \sqrt{n} b^k |c| b^m \geq |x|$ on the chart (`gcongr` from $u \leq b$). The summed bound $e^B u^h e(u)$ is continuous, hence integrable on the bounded chart (`Continuous.integrableOn_Ioc`). The second theorem splits each chart term as half-line minus tail (`setIntegral_union` on $(0, b] \cup (b, \infty)$) and evaluates the half-line by `standardIntegral1D_insertion` at $h + mp$.

4 Roadblocks and resolutions

- A `let-bound` integrand family F folds the `set` variable e out of sight: `rw [he]` then fails on goals like `Continuous (F p)`. Expose the body with `change` (the style linter rejects goal-changing `show`) before rewriting, or use `simp only [he]`, which also beta-reduces the exposed `lambda`.
- An unguided `rw [mul_assoc]` reassociated the *exponent* $-\beta n u^{2k}$ instead of the outer product; give the first factor explicitly, `mul_assoc (u ^ h)`.
- Pointwise congruence of a `HasSum` is cleanest as a function-level `rw` with a `funext` of the pointwise identity, rather than hunting for a `congr_fun` lemma.
- $0 \leq n$ is not needed for the chart theorem: $\sqrt{n}$ is total and the bound only uses `Real.sqrt_nonneg`; the hypothesis was dropped.

5 What was Mathlib, what was new

Mathlib: `hasSum_integral_of_dominated_convergence`, the exponential series, `Real.summable_pow_div_factorial`, `Continuous.integrableOn_Ioc`, `setIntegral_union`.
 New: the exact monomial tree on the chart and its rewriting in fluctuation functions with the $\Lambda(h, k)$ exponents visible.

6 Lessons

The paper's tree is, term by term, a reindexing of two facts already formalised (the kernel identity and the insertion identity) plus one series interchange; once the interchange is set up on the bounded chart the whole expansion is exact, and only the passage to the half-line introduces the $e^{-\varepsilon n}$ error. Generalising from a monomial to an analytic ξ is where the multi-index combinatorics enters.

7 Follow-ups

The 1D analytic- ξ tree (Cauchy product of the series for $(\xi(u) - \xi_0)^p$), the Mellin/state-density formulation, and the multivariate version remain.